

CYSTATIN C

An Effective Measurement Alternative for eGFR

RP Price: RM68 (Tag-on), RM100 (Single) *validity till 30/6/2026*

Code: GFRCY Sample: 3ml Plain



- The best indicator of overall kidney health and function is glomerular filtration rate (GFR).
- Creatinine, the most common serum marker used to assess GFR, can present problems due to variability in creatinine formation among individuals as well as tubular secretion.
- An alternative way to accurately assess GFR is by additionally measuring serum cystatin C.

CYSTATIN C is similar to creatinine in that it is produced at a constant rate and is freely filtered by the glomerulus. However, unlike creatinine, cystatin C is not solely produced in muscles and consequently, not linked to skeletal muscle mass. As a result, serum cystatin C correlates more closely to GFR than serum creatinine, especially in patients with atypical muscle mass. Race does not appear to greatly impact cystatin C serum concentrations. Thus, race isn't required for calculating an estimated GFR using cystatin C.

Advantages

- Provides an index of glomerular filtration rate.
- Estimates glomerular filtration rate in patients presenting with:
 - Abnormal muscle mass (e.g., obesity and malnourishment).
 - Rapidly changing muscle mass (e.g., hospitalized patients).
- Confirms or refutes an abnormal creatinine-estimated GFR.

Classification of Chronic Kidney Disease

Stage	GRF	Description	Cystatin C (mg/L)
1	90+	Normal or High GFR	≤ 0.85
2	60-89	Mild decreased in GFR	0.86-1.25
3	30-59	Moderately decreased in GFR	1.26-2.34
4	15-29	Severe decreased in GFR	2.35-4.16
5	<15	Kidney Failure	4.16

Source: <https://news.mayocliniclabs.com/renal/kidney-function/cystatin-c/>

